SEX-RELATED DIFFERENCES IN THE PHENOTYPIC EXPRESSION OF THE APOLIPOPROTEIN A-I_{Milano} MUTANT

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Apolipoprotein A-I_{Milano} (apo A-I_M) is a molecular variant of apo A-I characterized by a cysteine for arginine substitution at the position 173 in the primary sequence of apo A-I (Franceschini et al, 1980; Weisgraber et al, 1980, 1983). The presence of a single free cysteine in the mutant apolipoprotein leads to the formation of disulfide-linked homodimers (A-I_M/A-I_M) and heterodimers with apo A-II (A-I_M/A-II). The presence of these high molecular weight forms of the variant, slowly catabolized *in vivo* (Gregg et al, 1989), confers an increased metabolic stability to the carriers' high density lipoproteins (HDL), both *in vitro* (Franceschini et al, 1990a) or *in vivo* (Calabresi et al, 1992b). In an *in vitro* system, the interconversion of A-I_M HDL is impaired vs the case of control HDL, possibly due to a decreased capacity to accept/exchange lipids with triglyceride-rich lipoproteins (TGRLP); this same effect is observed *in vivo*, during the postprandial phase, when a pronounced postprandial rise of plasma and TGRLP triglyceride levels does not result in a parallel increase in the transfer of TGRLP triglycerides to HDL.

Thirty-six carriers of the A-I_M variant, aged between 1 and 79 years and all heterozygotes for the mutant apolipoprotein, have been identified, all descending from a single couple who lived in a small village in Northern Italy in the second half of XVIII century (Gualandri et al, 1985). In contrast to other recognized molecular variants of apo A-I (von Eckardstein et al, 1990), with the possible exception of the Pro_{165}→Arg mutant (von Eckardstein et al, 1989), the presence of the apo A-I_M variant has a significant effect on lipoprotein structure/function. The A-I_M carriers show multiple abnormalities in the lipid/lipoprotein pattern, mainly affecting the high density lipoprotein system (Franceschini et al, 1980, 1985; Gualandri et al, 1985): the plasma levels of HDL are constantly reduced compared to control subjects. Heterogeneous abnormal HDL_3, generally small in size and with a low flotation rate, are present in carriers' plasma, while plasma HDL_2 levels are markedly reduced (Franceschini et al,

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1982, 1987). In addition to the abnormalities in the HDL fraction, the A-I_M carriers show a higher prevalence of hypertriglyceridemia compared to close, non-affected relatives (Franceschini et al, 1985; Gualandri et al, 1985). The intriguing aspect in this syndrome is that, in spite of an unfavorable lipid/lipoprotein profile (Manninen et al, 1992) the apo A-I_M carriers have a very low incidence of coronary artery disease (Gualandri et al, 1985).

In spite of a well defined, typical lipid/lipoprotein pattern in the A-I_M carriers, a significant variability has been detected among the individual carriers (Franceschini et al, 1981, 1985). In particular, the HDL-cholesterol (HDL-C) levels are distributed over a relatively wide range (6-35 mg/dl) and are closely related to other lipoprotein parameters, such as the plasma triglyceride levels and the HDL particle size distribution (Franceschini et al, 1987). To gain further insight into the mechanisms responsible for this heterogeneity in the phenotypic expression of the mutant apolipoprotein, we examined the sex-related differences in the lipid/lipoprotein profile and possible correlations between the major lipid/lipoprotein parameters, with particular attention to the ratio between the mutant and the normal apolipoprotein.

Sex-related differences in the lipid/lipoprotein pattern in the A-I_M carriers

In this study we examined the plasma lipid/lipoprotein levels in 28 adult A-I_M carriers, 12 females and 16 males. For most of the examined parameters a mean value, calculated from at least 3 measurements in the last three years, was considered for each individual carrier. The difference between females and males was tested by one-way ANOVA; a probability value of less than 0.05 (two-tailed) was considered as significant.

The anthropometric and lipid/lipoprotein levels in the examined subjects are reported in Table 1. The two groups did not differ for age and BMI. The only significant differences between females and males were in the HDL levels. Very similar mean values were found for plasma total and low density lipoprotein (LDL) cholesterol, and apo B levels. LDL particles of normal size were found in both females and males. Plasma triglyceride levels were higher compared to nonaffected subjects from the same families (Franceschini et al, 1985; Gualandri et al, 1985), but very similar between female and male carriers. Cholesterol esterification, evaluated as cholesterol esterification rate and plasma LCAT concentration (Table 1), was similarly impaired

in both sexes compared to controls (Franceschini et al, 1990b). Plasma CETP levels, measured in a small subset of carriers, were also similar in the two groups.

Table 1. Anthropometric and lipid/lipoprotein parameters in female (n=12) and male (n=16) A-I_M carriers (mean ± SEM).

	Females	Males
Age (y)	43.8 ± 4.9	44.8 ± 4.8
BMI (kg/m²)	24.2 ± 0.8	24.1 ± 0.8
Total cholesterol (mg/dl)	189.9 ± 10.0	180.7 ± 11.8
Free cholesterol (%)	33.1 ± 1.0	33.3 ± 0.7
Triglycerides (mg/dl)	188.8 ± 21.3	175.6 ± 20.2
LDL-cholesterol (mg/dl)	130.9 ± 7.6	130.1 ± 8.8
LDL size (nm)	25.5 ± 0.2	25.5 ± 0.1
Apolipoprotein B (mg/dl)	108.3 ± 8.8	102.4 ± 7.3
HDL cholesterol (mg/dl)	21.3 ± 2.1	15.5 ± 1.6*
HDL_2-cholesterol (mg/dl)	3.6 ± 0.5	2.3 ± 0.3*
HDL_3-cholesterol (mg/dl)	17.7 ± 1.7	13.2 ± 1.3*
Apolipoprotein A-I (mg/dl)	94.3 ± 6.8	70.5 ± 6.7*
Apolipoprotein A-II (mg/dl)	24.8 ± 1.8	21.2 ± 1.3
Apolipoprotein $A\text{-}I_M/A\text{-}I_N$	1.03 ± 0.07	2.00 ± 0.22*
HDL particle score (AU)	2.48 ± 0.14	2.82 ± 0.11*
CER (nmol/ml/h)	23.7 ± 4.1	19.6 ± 1.5
LCAT (μg(ml)	3.9 ± 0.7 (n=7)	3.0 ± 0.1 (n=10)
CETP (μg/ml)	1.0 ± 0.1 (n=3)	0.9 ± 0.1 (n=3)

* significantly different from females

By contrast, HDL-C was 27% lower in males, due to a reduction of both HDL_2-C and HDL_3-C, by 36% and 25%, respectively. Plasma apo A-I levels, evaluated by immunoturbidimetry, which estimates the concentration of total apo A-I (normal A-I plus the various molecular forms of the mutant) (Weisgraber et al, 1980), were also significantly lower in males (-25%), while the reduction of apo A-II (-15%) did not reach significance ($p=0.12$); due to these changes, the A-I/A-II ratio was significantly lower in males (3.36 ± 0.27 vs 3.85 ± 0.16), possibly reflecting a more pronounced reduction of plasma Lp A-I than Lp A-I:A-II (Bekaert et al, 1992).

A dramatic difference between female and male carriers was detected in the mutant/normal apolipoprotein ratio, analyzed by isoelectric focusing under reducing conditions (Franceschini et al, 1981); in females the contribution of the two apolipoproteins to total plasma A-I was nearly identical, while males had a two-fold higher concentration of the mutant vs normal apo A-I (Table 1). This highly significant difference ($p=0.0014$) is probably responsible for the pronounced dissimilarity in the HDL particle size distribution between females and males. We previously showed that the abnormal HDL particle distribution in the carriers can be differentiated into three HDL patterns (Franceschini et al, 1987): pattern I is the most abnormal, characterized by a predominance of small HDL_{3b} particles; in pattern III, defined by a relatively normal HDL distribution, HDL_{3a} are the major component; the intermediate pattern II is characterized by a nearly equivalent contribution by HDL_{3a} and HDL_{3b}. Pattern I was significantly more frequent among males, while most of female carriers had a relatively normal HDL particle distribution (Pattern III, Table 2).

Table 2. Individuals with different HDL patterns among female and male $A\text{-}I_M$ carriers.

	Females	Males
Pattern I	3 (25.0%)	10 (62.5%)
Pattern II	2 (16.7%)	3 (18.7%)
Pattern III	7 (58.3%)	3 (18.7%)

HDL patterns have been described in Franceschini et al, 1987.

When the entire HDL particle distribution was evaluated by assigning each subject an HDL particle score (Calabresi et al, 1992a), so that a higher value reflects a distribution shifted toward particles of smaller size, a significantly lower mean value was found in males vs females (Table 1).

Since the post-menopausal age is associated with significant changes in lipid/lipoprotein metabolism (Seed, 1991), pre- (n=8) and post-menopausal (n=4) female carriers were analyzed separately. The only significant differences between the two groups were in plasma total and LDL cholesterol levels, both higher in post-menopausal (total: 224.2±8.2 mg/dl; LDL: 153.3±9.6 mg/dl) vs pre-menopausal women (172.7±8.9 mg/dl and 119.7±6.1 mg/dl, respectively). Differently from what observed in non carrier females (Masarei et al, 1980), post-menopausal carriers showed higher plasma HDL levels vs pre-menopausal women (Table 3), although only the difference in HDL_2-C reached borderline statistical significance ($p=0.057$). The A-I_M/A-I apolipoprotein ratio was almost identical in the two groups (pre: 1.01±0.07; post: 1.08±0.11).

Table 3. Plasma HDL levels (mg/dl) in pre-menopausal (n=8) and post-menopausal (n=4) A-I_M carriers

	Pre-menopausal	Post-menopausal
HDL-cholesterol	19.4 ± 1.6	25.0 ± 3.6
HDL_2-cholesterol	3.1 ± 0.3	4.9 ± 0.8
HDL_3-cholesterol	17.4 ± 1.7	20.8 ± 2.4
Apolipoprotein A-I	87.2 ± 6.1	108.2 ± 9.8
Apolipoprotein A-II	23.2 ± 1.5	27.7 ± 2.7

In summary, the male carriers have a higher mutant/normal apo A-I ratio, lower apo A-I and HDL-C levels, and smaller HDL particles, compared to females, both pre- or post-menopausal. To investigate the role of the A-I_M/A-I ratio in determining the alterations in the HDL system typically found in the A-I_M carriers, a series of correlations was calculated.

Relationship between the A-I_M/A-I ratio and lipid/lipoprotein parameters

The results of simple correlation analyses performed on the data from the whole series of 28 subjects are reported in Table 4. The apo A-I_M/A-I ratio was strongly correlated with several HDL parameters, including HDL-C, HDL_2-C, HDL_3-C and apo A-I levels (Figure 1); the correlation with plasma apo A-II levels was weaker, but still statistically significant.

Table 4. Correlations between the mutant/normal apo A-I ratio and lipid/lipoprotein parameters in 28 A-I_M carriers.

	r	p
HDL-cholesterol	- 0.60	0.0007
HDL_2-cholesterol	- 0.57	0.0017
HDL_3-cholesterol	- 0.68	< 0.0001
Apolipoprotein A-I	- 0.72	< 0.0001
Apolipoprotein A-II	- 0.41	0.031
HDL particle score	0.76	< 0.0001
HDL pattern	- 0.64	0.0003

The mutant to normal apo A-I ratio was also related to the type of HDL particles present in the individual's plasma, as demonstrated by the negative correlation with the HDL pattern (I to III) and by the positive correlation with the HDL particle score (the higher the score, the smaller the particles). The correlations with other lipid/lipoprotein, or enzymatic parameters were non significant, except for a weak, positive correlation with plasma triglyceride levels ($r=0.44$, $p<0.05$)

Conclusions

The present study revealed a characteristic sex-related difference in the concentration of various HDL parameters in the A-I_M carriers. Females had significantly higher

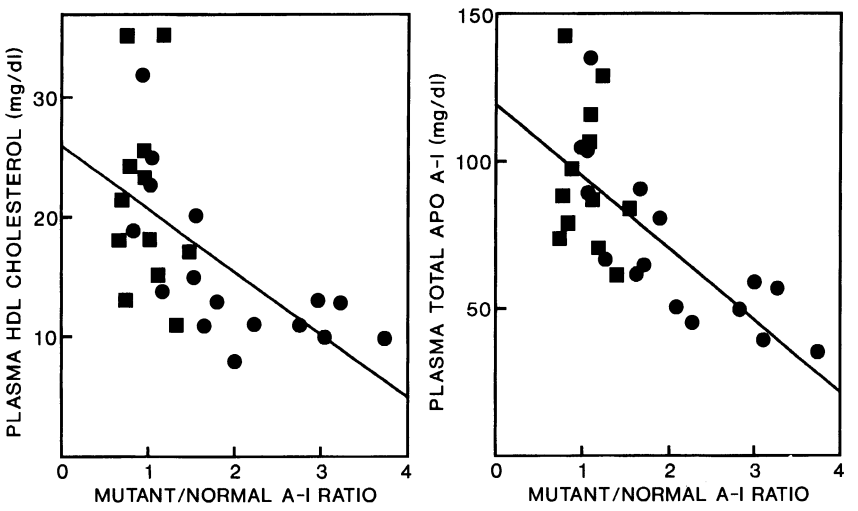


Figure 1. Relationship between the mutant/normal apo A-I ratio and plasma HDL-C and total A-I levels in 28 A-I_M carriers, 12 females (■) and 16 males (●).

HDL-C and apo A-I levels, and larger HDL particles than males. These sex-related differences are consistent with those found in normal individuals (Albers et al, 1976), but seem to be amplified in the A-I_M carriers, possibly due to the common, well defined genetic background of these subjects, all belonging to the same large kindred (Gualandri et al, 1985). The mechanism responsible for the higher HDL levels usually seen in women is generally thought to be related to the inhibitory action of estrogens on hepatic lipase (HL)(Tikkanen et al, 1982), leading to a reduced turnover of HDL; the dissociation of apo A-I from HDL is prevented (Clay et al, 1991), further contributing to the elevated plasma apo A-I levels (Goldberg et al, 1990). By contrast, the increased HL activity in males results in lower HDL levels vs females. The hypothesis of a regulatory role of sex hormones on plasma HDL levels is confirmed by epidemiological data showing significant reductions of HDL in post-menopausal women (Seed, 1991). This seems not to be the case in the A-I_M condition; post-menopausal females had the highest plasma HDL-C and apo A-I levels, while displaying the expected increase in plasma LDL-C vs pre-menopausal women.

The main finding of the present study was the disclosure of the strong effect of the mutant/normal apo A-I ratio on the HDl system in the individual carriers. A high var-

iability in this ratio was already revealed in the original A-I_M family (Franceschini et al, 1981) and now confirmed by the investigation of a larger number of A-I_M carriers; it is of interest to note that this variability was significantly higher among male (CV=44%) vs female (CV=23%) carriers (Fig. 1). The mutant/normal apo A-I ratio was strongly correlated with several HDL parameters, including HDL-C and apo A-I levels, HDL subfraction distribution and particle size. In male carriers, a relative abundance of the mutant apolipoprotein, which is catabolized faster than normal apo A-I (Gregg et al, 1989), would lead to low plasma levels of total A-I at steady state; the formation of apo A-I_M dimers should increase, and abnormal HDL, enriched in A-I_M dimers, should accumulate in plasma. By contrast, in female carriers, the predominance of normal apo A-I, with a slower catabolism vs the mutant, would result in higher, closer to normal apo A-I levels and HDL particle distribution. The minimal, non-significant sex-related difference in plasma apo A-II levels, together with the weaker correlation between the mutant/normal apo A-I ratio and apo A-II, vs A-I, levels indicate that the metabolism of apo A-II is less affected by the presence of the apo A-I mutant. This is consistent with the recently described minor sex-related difference in Lp-A-I:A-II levels compared to a marked decrease of Lp-A-I in male vs female carriers (Bekaert et al, 1992). All together these findings indicate that the individual capacity to regulate the proportion of mutant vs normal apo A-I plays a major role in determining the individual HDL pattern, and that sex, by unknown mechanisms and in addition to other factors, particularly in males, significantly contributes to this regulation.

A second major finding of this study was that, in spite of the 25-30% lower plasma HDL-C and apo A-I levels in male vs female carriers, plasma triglyceride concentrations were similarly elevated in both sexes. Moreover, and differently from normal subjects, no correlation could be found between fasting triglyceride and HDL-C levels, either in the whole series, or by separate analysis of female and male data. A completely different situation was found in the postprandial state, where the A-I_M carriers showed a three-fold higher postprandial lipemia, which was inversely correlated, similarly to controls, with fasting plasma HDL-C and apo A-I levels (Calabresi et al, 1992). These findings argue for a different co-regulation of plasma HDL and triglyceride levels in the fasting and postprandial state, in the A-I_M carriers. In the fasting state plasma HDL levels are mainly regulated by changes in the protein composition of HDL, and not by the lipid content, as occurs in normal subjects (Brinton et al, 1991). In the postprandial state, the presence of a reduced plasma pool of HDL, already enriched in triglycerides, leads to an impaired transfer/exchange of triglyceride between TGLRP and HDL, with a progressive accumulation of triglyceride

enriched particles in plasma (Calabresi et al, 1992). A contrasting situation occurs in normal subjects, particularly in those with coronary disease (Groot et al, 1991), where an increased postprandial lipemia, due to a defective lipolysis, results in low plasma HDL levels, possibly as a result of increased triglyceride transfer to HDL_2 with subsequent hydrolysis by hepatic lipase (Patsch et al, 1984). In the A-I_M syndrome, a primary defect in HDL structure leads to low plasma HDL levels and reduced triglyceride transfer to HDL; this condition is associated with elevated fasting and postprandial triglyceride level, but apparently good vascular conditions.

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